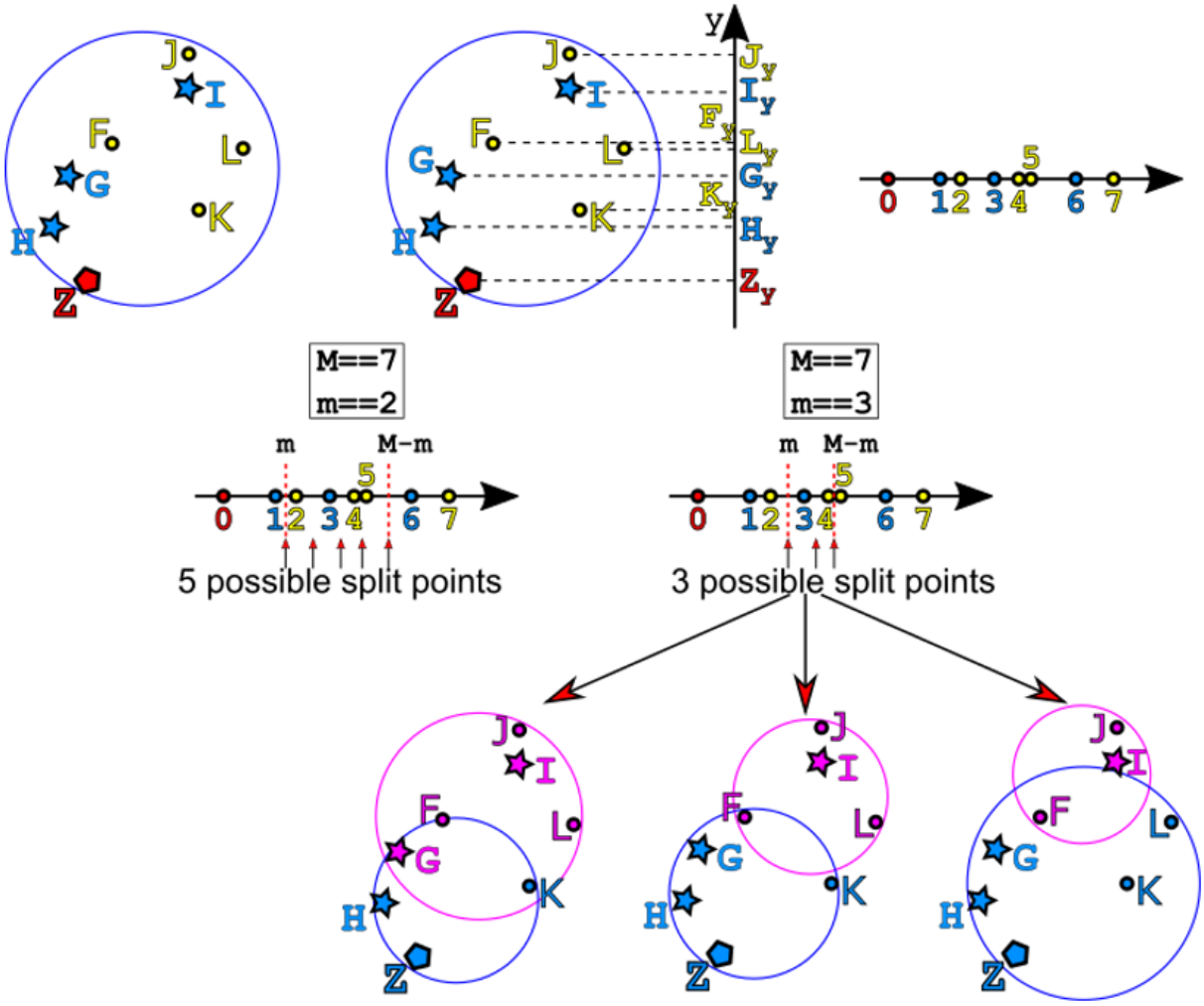


Figure 10.18 Splitting a set of point along the direction of maximum variance.
(Top) The bounding envelope and its points to split; the direction of maximum variance is along the `y` axis (center) so we project all points on this axis; on the right, we rotate the axis for convenience, and replace the point labels with indices.
(Middle) Given there are 8 points, we can infer `M` must be equal to `7`. Then `m` can be any value ≤3; since the algorithm chooses a single split index, partitioning the points on its two sides, and each partition need to have at least `m` points, depending on the actual value